

EndRing PKE candidate funnel

GOAL-SSI-2fa82a / BATCH-65c2dd

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Result

This batch implemented an explicit **propose** -> **evaluate** -> **gap-feed** -> **propose** -> **evaluate** research loop for public-key encryption motivated by the supersingular endomorphism-ring problem. It generated three candidate packets, froze both review inputs, ran independent obligation-based reviews, and produced **zero finalists**.

No secure PKE, EndRing security reduction, novelty result, implementation result, parameter set, or universal impossibility theorem was established. All work was symbolic: zero scientific, implementation, formalizer, benchmark, or parameter-selection runs were performed. The evidence supports scoped candidate findings only. **Method** [Final]

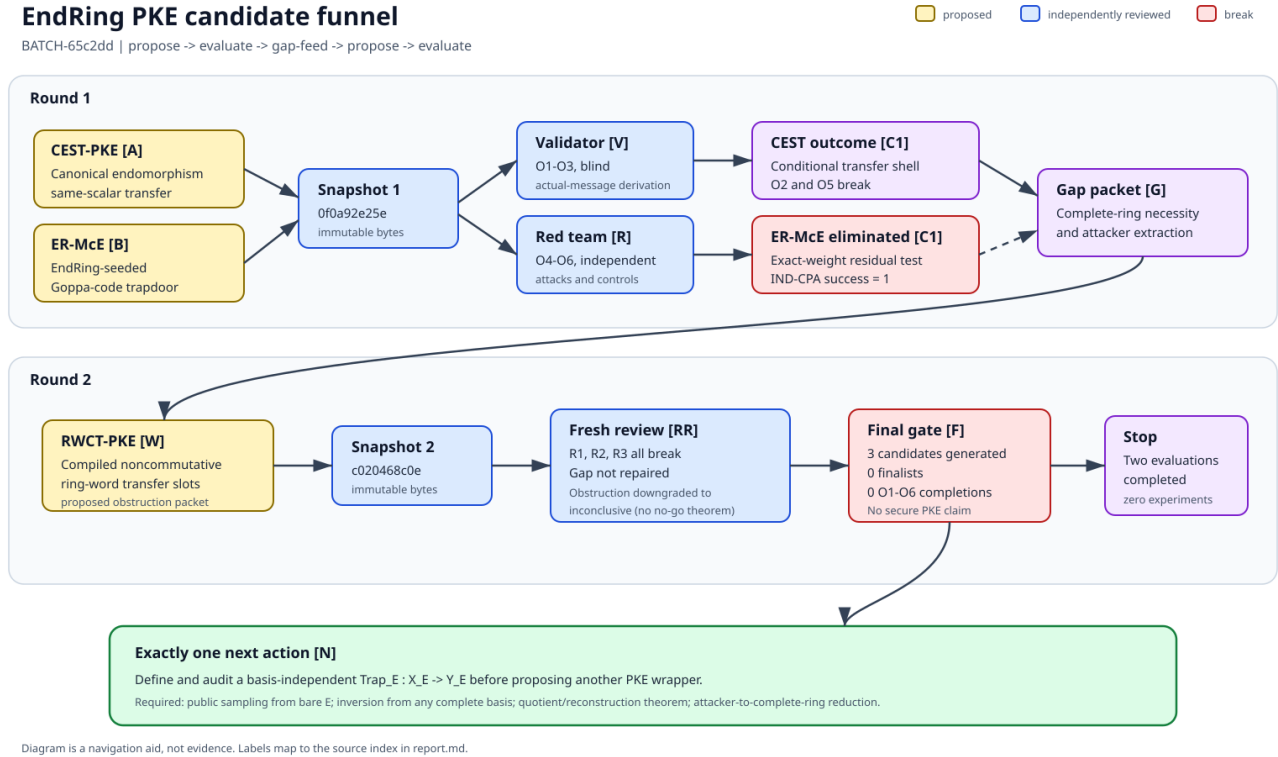


Figure 1: Candidate funnel. Proposed material is yellow, independent review is blue, breaks are red, and the next action is green. The diagram is a navigation aid, not evidence.

Method

The frozen methodology requires materially different recipient mechanisms, complete PKE algorithms and encodings, immutable snapshots before review, joint ownership rather than whole-claim voting, known-false controls, and one

structured gap-fed revision. A KEM-only result does not satisfy the PKE goal. The batch stops after the second evaluation. [Method](#)

Every candidate was judged against six obligations:

Gate	Question
O1	Is the exact EndRing problem, distribution, auxiliary view, representation, and setup specified?
O2	Are KeyGen, Enc, and Dec honest, typed, public/secret separated, and polynomial-time?
O3	Does Dec recover the selected plaintext for quantified keys, messages, coins, and failures?
O4	What secret-dependent recipient operation meets the public sender operation, and why is it not public?
O5	Does a scheme attacker yield a solver for the exact EndRing challenge in the correct direction?
O6	Is the complete adversarial view covered, including encodings, malformed inputs, repetition, and auxiliary data?

The controls included public recovery, leaked-plaintext known-false, wrong-key/malformed ciphertext, real-distribution embedding, actual-message recovery, and regression against previously failed lift/evaluator mechanisms. [Plan1] [Plan2]

Round 1: two independent candidates

Candidate A: CEST-PKE

Proposed mechanism. A complete evaluable witness for $\text{End}(\mathbf{E})$ selects an endomorphism $\mathbf{psi_E}$. The public key contains $\mathbf{P}$ and $\mathbf{A} = \mathbf{psi_E}(\mathbf{P})$. Encryption samples $\mathbf{r}$, publishes $\mathbf{R} = [\mathbf{r}]\mathbf{P}$, and masks the message with a hash of $[\mathbf{r}]\mathbf{A}$. Decryption computes $\mathbf{psi_E}(\mathbf{R})$. [A]

Derived correctness. For an accepted valid key,

$$\mathbf{psi_E}(\mathbf{R}) = \mathbf{psi_E}([\mathbf{r}]\mathbf{P}) = [\mathbf{r}]\mathbf{psi_E}(\mathbf{P}) = [\mathbf{r}]\mathbf{A}.$$

The blind validator independently rederived exact selected-message recovery. However, O2 breaks because the curve sampler, complete-ring verifier, canonical endomorphism selector, evaluator bounds, and retry bounds are not available. [V]

Security finding. The red team found no public polynomial transfer algorithm, so recipient functionality was not directly broken. But decryption uses only one $\mathbf{psi_E}$ evaluator: replacing the complete ring witness with that evaluator leaves every decryption transcript unchanged. The direct security object is therefore a hashed same-scalar-transfer problem, not the complete EndRing problem. [R]

The known EndRing implication has the wrong direction:

complete EndRing solver $\rightarrow$ derive $\mathbf{psi_E} \rightarrow$ compute transfer $\rightarrow$ decrypt.

No reduction embeds a bare curve-only EndRing challenge into the real public key and converts an IND-CPA attacker into a complete evaluable rank-four basis. O5 therefore breaks. The surviving statement is only a conditional transfer shell, not an EndRing-secure PKE. [R] [C1]

Candidate B: ER-McE

Proposed mechanism. A canonical encoding of a complete EndRing witness seeds a deterministic Goppa-code trapdoor. The public operation is raw McEliece encryption

$$\mathbf{c} = \mathbf{m} \mathbf{G_pub} + \mathbf{e}, \quad \text{wt}(\mathbf{e}) = \mathbf{t},$$

and the recipient uses the secret Goppa decoder. [B]

Decisive break. For distinct challenge messages m_0, m_1 , compute

$$x_i = c - m_i G_{\text{pub}}.$$

The true residual has weight exactly t . If both residuals had weight t , then the nonzero codeword $(m_0 - m_1)G_{\text{pub}} = x_0 - x_1$ would have weight at most $2t$, contradicting the declared minimum distance at least $2t+1$. Exactly one residual has weight t ; the challenge bit is recovered with probability one, for advantage $1/2$ under the packet's convention. This is a static symbolic counterexample and required no experiment. [R]

ER-McE is eliminated in its exact syntax. The result does not break general McEliece, EndRing, or every EndRing-seeded composition. Independently, the real public matrix cannot be generated from a bare EndRing challenge without the hidden ring seed, and code decoding remains a separate assumption. [R]

Composition and gap selection

The first composition copied each joint owner's verdict without voting. CEST-PKE had a conditional message-transfer identity but broke O2 and O5. ER-McE broke all five review joints and was eliminated by the public exact-weight test. No candidate was a finalist. [C1]

One decision-changing gap was selected:

Make complete EndRing information both sufficient and demonstrably necessary for recipient functionality on the exact real public-key distribution, and give the correctly directed IND-attacker-to-complete-ring reduction.

The gap packet forbade renaming transfer, CDH, torsion-action, or decoding hardness as EndRing; assuming a missing lift/evaluator; weakening selected-message IND-CPA; or treating KeyGen provenance as proof that the full ring is necessary. [G]

Round 2: gap-fed revision

RWCT-PKE

The fresh producer saw only the gap packet and contracts. It attempted to replace the single evaluator with polynomially many noncommutative words $w_j(B)$, each syntactically involving all four elements of a planted complete basis B . The public key contains $A_j = w_j(B)(P_j)$; encryption uses $[r_j]A_j$; decryption evaluates $w_j(B)([r_j]P_j)$. [W]

For a supplied valid planted tuple, the conditional identity

$$w_j(B)([r_j]P_j) = [r_j]w_j(B)(P_j) = [r_j]A_j$$

recovers the selected message. The producer nevertheless returned a scoped obstruction packet: decryption factors through the finite restricted evaluators, a bare E cannot generate the correlated public images, and an arbitrary EndRing basis requires an unavailable alignment to the planted word variables. [W]

Fresh review

The second packet was frozen at commit `c020468c0e` and evaluated by a fresh reviewer who did not read the first-round reports. All three revision joints break; the selected gap is not repaired. [S2] [RR]

The reviewer found:

1. **Ill-typed decryption.** KeyGen returns $sk=B$, while $Dec(sk, ct)$ reads pp, pk , word descriptors, points, and transcript domains that are neither in sk nor passed as inputs.
2. **Ambiguous honest zero coins.** Uniform r_j includes zero, so honest ciphertexts can contain the identity. Correctness requires subgroup membership to accept it, whereas an exact-order- N_j point test would reject.
3. **Syntactic dependence is not semantic dependence.** The allowed word $X_1 + (X_2 - X_2) + (X_3 - X_3) + (X_4 - X_4)$ mentions every variable but collapses to X_1 , replaying the single-evaluator route.

4. **The three reduction arrows remain absent.** Bare **E** does not sample the real key, an attacker bit does not yield a complete ring, and an arbitrary complete basis is not aligned with the planted variables.

The factorization through restricted evaluators is exact, but it does not prove a family-level no-go result. No strict efficient quotient, collision pair, meta-reduction, or reconstruction impossibility was supplied. Accordingly, the producer’s proposed scoped obstruction is **inconclusive**, not an impossibility theorem. [RR]

Final scorecard

Candidate	O1	O2	O3	O4	O5	O6	Disposition
CEST-PKE	inconclusive	breaks	conditional identity verified	inconclusive	breaks	inconclusive	transfer shell only
ER-McE	breaks	breaks	incomplete conditional claim	breaks	breaks	breaks	eliminated by perfect IND-CPA distinguisher
RWCT-PKE	open	open	conditional identity only	open	open / missing reduction	open	selected gap not repaired

The batch generated three candidate packets, independently evaluated all three, and found zero candidates with complete O1-O6 discharges. No implementation or parameter run is warranted for these exact syntaxes. [Final]

What the loop learned

- A complete EndRing witness used during KeyGen proves sufficiency only. It does not make the complete ring necessary for decryption or extractable from an attacker.
- Every public-key field correlated with the hidden witness must be sampled from the exact EndRing challenge or justified by an explicit distribution compiler.
- The reduction output must be a complete evaluable rank-four basis, not one evaluator, a transfer point, torsion action, decoder, endpoint, or planted label.
- Syntactic use of four basis variables can cancel. Semantic dependence needs a non-factorization or reconstruction theorem.
- Elementary selected-message attacks belong before assumption-level analysis; this exposed ER-McE immediately. [Final]

Exactly one next action

Before proposing another encryption wrapper, define and audit one basis-independent trapdoor map

$\text{Trap}_E : X_E \rightarrow Y_E$

with all four properties:

1. public instance sampling and forward evaluation are polynomial from a bare **E** drawn from the exact EndRing challenge distribution;
2. any complete evaluable basis of $\text{End}(E)$, not one planted basis, enables polynomial inversion;
3. inversion either provably does not factor through a smaller witness, or a reconstruction theorem recovers the complete ring from that witness; and
4. a selected-message IND attacker yields a complete evaluable basis with explicit game hops, oracle simulation, loss, runtime, and abort bounds.

If any property is unavailable, the next record should preserve that exact failed arrow before another PKE syntax is built. The goal remains active; this batch closes as complete with no finalist. [Final]

Evidence and reproducibility

- Initial producer bytes were frozen at commit `0f0a92e25e`; the revision bytes were frozen at `c020468c0e`. [S1] [S2]
- The initial independence check passed for two mutually blind reports; the revision independence check passed for the fresh report. [V] [R] [RR]
- The repository ledger validator returned the inherited baseline of 105 diagnostics and 1,210 grandfathered legacy diagnostics, with no batch-local diagnostic after metadata correction.
- No primary paper was read in this batch. Prior-art comparisons used only abstract-level repository pointers, so novelty remains unverified. [R]
- No diagram or PDF is evidence; scientific statements above cite the underlying candidate, review, or composition artifact.

Source index

- [Method](#) [Frozen methodology](#)
- [A] [CEST-PKE candidate](#)
- [B] [ER-McE candidate](#)
- [S1] [Initial snapshot receipt](#)
- [Plan1] [Initial review plan](#)
- [V] [Blind O1-O3 validation](#)
- [R] [Initial O4-O6 red-team report](#)
- [C1] [Initial composition](#)
- [G] [Structured gap packet](#)
- [W] [RWCT-PKE revision](#)
- [S2] [Revision snapshot receipt](#)
- [Plan2] [Revision review plan](#)
- [RR] [Fresh revision review](#)
- [Final] [Final composition](#)